MOBILE APPLICATION DEVELOPMENT - MCA 2027

1. Design and develop an Android application featuring three activities. The first activity must provide a secure login gateway for both Students and Quiz Creators. The second activity allows Creators to dynamically compile quiz questions. The final activity enables Students to attempt the quiz, where the system must automatically evaluate answers and compute scores. The final marks must persist securely in the application across user logout sessions.
2. Implement an Android application that uses SharedPreferences to store and manage student profiles, including their Name, Internal Marks, and External Marks. In a dedicated "View Marks" activity, develop a user-friendly interface that populates these records dynamically, allowing a user to retrieve and view a specific student's marks by selecting their profile from either a Spinner or an AutoCompleteTextView
3. Develop a grocery billing application designed to generate dynamic invoices. The primary activity must allow users to select items from a predefined list and input quantities. Both the item catalog and their respective unit prices must be saved using SharedPreferences. Upon triggering the "Generate Bill" action, the application must compute the total cost and display a itemized breakdown—detailing item names, quantities, individual prices, and the final grand total—within a scrollable container (ScrollView).
4. Develop a secure Pizza Shop application featuring a functional Login activity. Upon successful authentication, navigate the user to a dashboard offering two options: "Customize Pizza" and "View Order." The customization activity must allow users to specify pizza size and base pricing via RadioGroup/RadioButton, select multiple toppings with individual pricing using CheckBox, and choose the crust type from a Spinner. The "View Order" interface must dynamically aggregate these selections and display the total calculated price.
5. Develop a Marks Card Generator application tailored for educators. The initialization activity must allow teachers to select a student ID from a Spinner and input marks for various subjects using validation-checked EditText fields. Upon selecting "Compile Report" from the Options Menu, the application must compute the total marks, percentage, and final grade, rendering a comprehensive, itemized report card within a structured TableLayout that is saved locally via SharedPreferences.
6. Design a digital Attendance Marking system consisting of separate workflows for instructors and students. The application must feature a ListView or RecyclerView loaded dynamically from SharedPreferences to display the student roster. Instructors must be able to mark status (Present/Absent) via toggle switches, while an Options Menu provides a "Generate Absentees Report" action. Student data and final attendance percentages must persist securely in SharedPreferences across user sessions.

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7. Implement a Faculty Leave Management application requiring user authentication. The primary dashboard must present a layout featuring an Options Menu with tabs for "Apply Leave," "Check Status," and "Leave Balance." The leave application form must utilize a DatePicker dialog to pick calendar dates and a Spinner for leave classification (Casual, Medical, Duty). Application statuses must be written to and updated dynamically within SharedPreferences, reflecting instantly on the status dashboard.

8. Develop an E-Library application that allows users to browse an institutional book catalog. The main interface must showcase books categorized inside a scrollable view. Long-pressing any book item must trigger a Context Menu with options like "View Details," "Reserve Book," and "Add to Wishlist." The application must track book availability and reserve items locally via SharedPreferences, dynamically updating the UI count when a book is checked out.

9. Design an application enabling students to anonymously lodge institutional grievances. The input form must feature a Spinner to categorize the issue (Academic, Infrastructure, Hostel) and a multi-line EditText for descriptions. Implement an Options Menu for administrators to sort tickets by priority or resolution status. All submitted cases must be logged into SharedPreferences using key-value indexing, persisting the ticket history along with time stamps and unique tracking IDs.

10. Develop a placement coordination application featuring dual roles for Students and Placement Officers. Students can view available job drives and apply using checkboxes to verify prerequisite criteria, while Placement Officers can post new drives via an Options Menu action. The app must store company profiles, eligibility cutoffs, and applicant numbers inside SharedPreferences, displaying live eligibility status alerts to students based on their profile metrics.

11. Design and develop a smart room climate and concierge configuration console. The primary panel activity outlines appliance toggles using a balanced grid layout matrix featuring lighting toggles, airconditioning control RadioButtons, and amenity request CheckBoxes. Validating the panel layout inputs registers the preference values inside Shared Preferences and executes an Explicit Intent launching the room services dashboard activity. This second activity dynamically parses the environment metrics to greet the guest and maps operational active room requests inside a scrollable ListView via an Android Adapter. Short-clicking item elements fires an informative confirmation Toast, whereas long-pressing row indices reveals a Context Menu executing an Implicit Intent web request detailing external hotel spa coordinates.

MOBILE APPLICATION DEVELOPMENT - MCA 2027

12. Design and develop a fitness tracking application. The first activity presents a profile setup wizard featuring an EditText for Username and Target Weight, requiring non-empty input validation. Clicking a "Setup Complete" button triggers an Alert Dialog; upon confirmation, the Username is saved into Shared Preferences, and an Explicit Intent opens the second activity. The second activity reads the username from preferences to build a dynamic banner, and loads available personal trainers into a ListView using an Android Adapter. Clicking a trainer item fires a Toast alert, whereas long-pressing the trainer reveals a Context Menu that executes an Implicit Intent to open the system email client to schedule a consultation session.

13. Design and develop an Android application for a "Patient Appointment & Clinic Dashboard" by creating two activities. The first activity must utilize a structured layout to display form widgets for capturing a patient's Full Name, Age, and Preferred Token/Date, incorporating runtime validation on all input fields and an Options Menu to clear the form data. Upon successful validation, the application must store the Patient's Name using Shared Preferences, bundle the remaining inputs, and transition to the second activity via an Explicit Intent. The second activity must dynamically greet the patient by retrieving their name from preferences and use an Android Adapter to render a collection of available specialist doctors inside a ListView. Clicking a doctor item must pop up an Alert Dialog that triggers a Toast notification upon confirmation, while long-pressing an item must open a Context Menu that fires an Implicit Intent to launch an external web browser or mapping application for hospital route navigation.

14. Design and develop an Android application for a "Vehicle Rental System & Drive Dashboard" by creating two activities. The first activity must utilize a structured layout to display form widgets for capturing a driver's Full Name, Driving License Number, and Mobile Number, incorporating runtime validation on all input fields and an Options Menu to clear the form data. Upon successful validation, the application must store the Driver's Name using Shared Preferences, bundle the remaining inputs, and transition to the second activity via an Explicit Intent. The second activity must dynamically greet the driver by retrieving their name from preferences and use an Android Adapter to render a collection of available car and bike categories inside a ListView. Clicking a vehicle item must pop up an Alert Dialog that triggers a Toast notification upon confirmation, while long-pressing an item must open a Context Menu that fires an Implicit Intent to launch an external web browser displaying terms of service and insurance policy rules.

MOBILE APPLICATION DEVELOPMENT - MCA 2027

15. Design and develop an Android application for an "Alumni Network & Placement Connect" by creating two activities. The first activity must utilize a structured layout to display form widgets for capturing an alumnus's Full Name, Graduation Year, and Current Company, incorporating runtime validation on all input fields and an Options Menu to clear the form data. Upon successful validation, the application must store the Alumnus's Name using Shared Preferences, bundle the remaining inputs, and transition to the second activity via an Explicit Intent. The second activity must dynamically greet the user by retrieving their name from preferences and use an Android Adapter to render a collection of mentorship tracks and networking chapters inside a ListView. Clicking a chapter item must pop up an Alert Dialog that triggers a Toast notification upon confirmation, while long-pressing an item must open a Context Menu that fires an Implicit Intent to open the device email application with the chapter lead's address pre-populated.